1. Clinical Question & Core Hypothesis

Clinical Question

Why do some localized prostate cancers with adverse histology (cribriform architecture) not progress unless they also activate specific transcriptomic metastatic programs, and conversely why some high-risk transcriptomic tumors do not rapidly progress unless accompanied by invasive morphologies? 2) Tier 1 paths used: (a) Cribriform architecture → morphological axis of collective invasion/structural routes for dissemination; (b) Decipher score → transcriptomic programs that predict metastatic competency. I built the hypothesis on the mechanistic intersection of these two paths (morphology enabling dissemination × transcriptional metastatic potential). 3) Tier 2 supporting evidence: published associations of cribriform pattern with worse outcomes and with invasive features; validation of Decipher as a predictor of metastasis/recurrence; biological plausibility that morphologic invasion and cell-intrinsic metastatic programs synergize. 4) Endpoint selection: PFI (progression-free interval) maps best to the clinical outcomes of interest here—biochemical recurrence and metastasis—and is sensitive to earlier clinical events that the biomarker aims to predict. 5) Deliberate exclusions: I excluded broader clinical covariates (PSA, treatment modalities) and multiple additional genomic modifiers from the core hypothesis to keep focus on the mechanistic interaction between morphology and transcriptomic metastatic programs; those factors are listed as secondary_modifiers for follow-up analysis?

Working Hypothesis

Cribriform architecture creates a dominant collective invasion axis that enables tumor cells to disseminate along defined morphological routes. High Decipher transcriptional programs encode metastatic competence, including motility, invasion, and extracellular matrix remodeling. I hypothesize a synergistic interaction: cribriform architecture is necessary to realize the full metastatic potential of tumors with high Decipher scores, such that tumors positive for cribriform pattern and with high Decipher scores exhibit a greater risk of progression to metastasis and shorter progression-free interval (PFI) than would be predicted by either feature alone. In non-cribriform tumors, even a high Decipher score may have limited impact on progression due to the absence of coherent invasion routes. The mechanistic basis for this interaction likely involves spatial proximity of Decipher-driven invasion programs to cribriform invasion fronts, enabling efficient cell shedding and colonization along collective migratory paths. This interaction should be detectable as a statistically significant Cribriform × Decipher interaction predicting PFI after adjusting for standard clinical covariates, and should be supported by (a) spatial transcriptomic mapping showing enrichment of Decipher-program genes at cribriform invasion fronts, (b) functional readouts in model systems where cribriform-like architecture and Decipher-encoded programs are concomitantly manipulated to alter invasion and metastatic potential. Collectively, these data would support a combined histology-transcriptomic biomarker that identifies patients at highest risk who may benefit from treatment intensification or intensified surveillance, beyond what either cribriform status or Decipher score provides alone. Specific testable predictions: (1) In cribriform-positive tumors, high Decipher score confers a significantly shorter PFI than cribriform-positive tumors with low Decipher; (2) In non-cribriform tumors, Decipher score stratifies risk to a lesser extent; (3) There is a statistically significant interaction term between cribriform status and Decipher score in Cox models predicting PFI; (4) Spatial transcriptomics demonstrates co-localization of Decipher-related transcripts with invasion-front markers specifically in cribriform tumors; (5) In organoid/animal models, inducing cribriform-like architecture together with high Decipher gene expression results in greater invasion and metastatic spread than either manipulation alone, while disrupting one component mitigates dissemination despite the other.

Element	Value
Imaging feature	Cribriform architecture
Anchor gene(s)	Decipher_score
Primary endpoint	PFI

2. Why This Gene — Not Others

Primary hypothesis gene(s)

Decipher_score: Selected as a core mechanistic driver from Tier 1 knowledge graph analysis.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	PTEN loss	Supporting modifier from knowledge graph
Secondary	TP53 mutation	Supporting modifier from knowledge graph
Secondary	Gleason/ISUP grade group	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: In retrospective cohort data, the prognostic benefit of the imaging feature will be modified by the anchor gene expression level (significant interaction term).

Prediction 2: At the tissue level, tumors with the imaging feature but low anchor gene activity will show immune cells confined to structural peripheries rather than infiltrating tumor nests.

Prediction 3: In functional experiments, restoring or enhancing the anchor gene's signaling will improve immune-mediated tumor control in imaging-feature-positive models.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.
- Multi-modal validation pipeline: computational -> spatial -> functional, providing mechanistic depth beyond association.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- PTEN loss: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- TP53 mutation: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.